

# HAT-P-32b

R-0268

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-32\_250914 · created 2026-07-17T12:44:48+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460932.8197 +/- 0.0041 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.181 +/- 0.016                 |
| Transit depth (Rp/Rs)^2                           | 3.29 +/- 0.56 [%]               |
| Orbital Inclination (inc)                         | 89.72 +/- 1.71                  |
| Airmass coefficient 1 (a1)                        | 1.011 +/- 0.024                 |
| Airmass coefficient 2 (a2)                        | -0.019 +/- 0.038                |
| Scatter in the residuals of the lightcurve fit is | 2.37 %                          |
| Best Comparison Star                              | #2 - [480, 419]                 |
| Optimal Aperture                                  | 4.32                            |
| Optimal Annulus                                   | 5.47                            |
| Transit Duration (day)                            | 0.1198 +/- 0.0044               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.181 ± 0.016 vs 0.1489 (deviation 2.01σ) |
| Duration fit vs published | 0.1198 d vs 0.1304 d (deviation 2.41σ)    |
| Mid-time O–C              | -2.9 min                                  |
| Depth SNR                 | 11.3                                      |
| Residual scatter          | 2.37 %                                    |

### Light curve

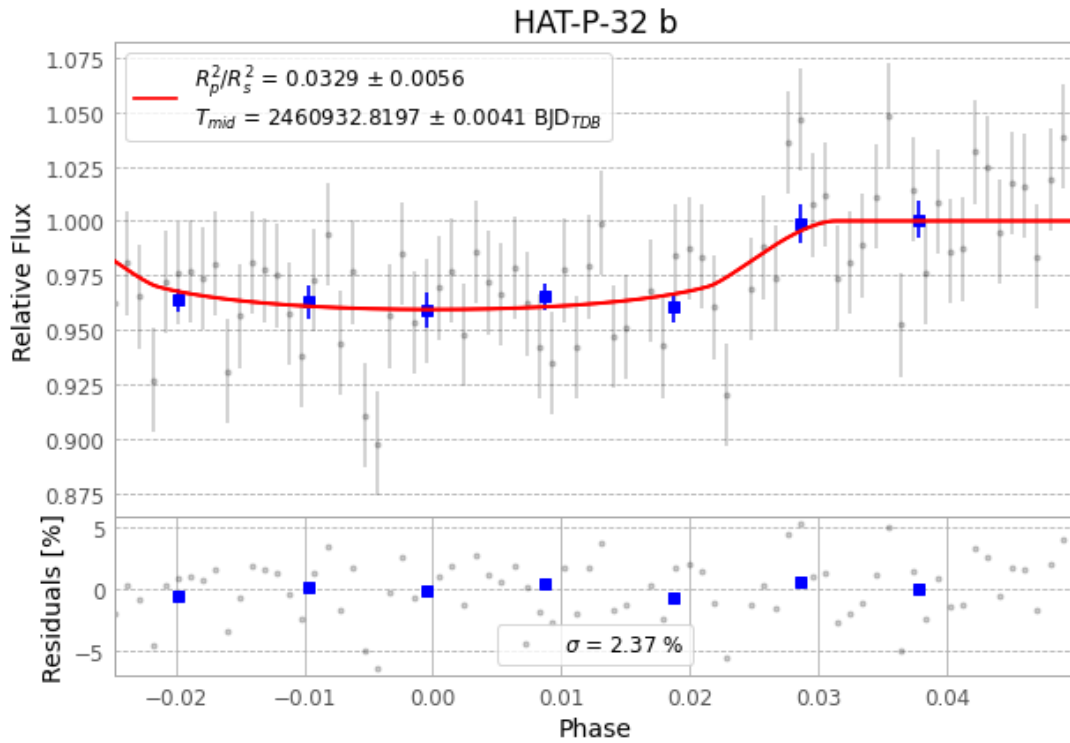


Figure 1 — FinalLightCurve\_HAT-P-32 b\_2025-09-13.png (SHA-256 4e22323025082d42...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 219], 2 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 b011d707defa8a2f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1cb2214

## Data provenance

79 FITS frames (52 MB), HATP-32250914061816.FITS → HATP-32250914101215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-250914.log (9f20a1b44d56...), FinalLightCurve\_HAT-P-32 b\_2025-09-13.png (4e2232302508...), FinalLightCurve\_HAT-P-32 b\_2025-09-13.csv (5dc00e1941d8...)

## Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 12:44Z.